

Soumik Mukhopadhyay

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RESEARCH INTEREST

My research interests lie in computer vision, particularly at the intersection of generative modeling, multimodal learning, representation learning, and efficient training. I study how structure across scales, learned representations, and generative dynamics can improve visual understanding and enable efficient image and video generation. I have prior experience in medical image analysis.

EXPERIENCE

UNIVERSITY OF MARYLAND, COLLEGE PARK RESEARCH ASSISTANT	May 2021 - Present
Forgery detection, Sparse 3D novel view synthesis	Prof. Abhinav Shrivastava Dept. of Computer Science
ADOBE, SAN JOSE RESEARCH SCIENTIST INTERN	May 2026 - Present
Pixel diffusion models	Aniruddha Mahapatra, Aseem Agarwala Video AI Lab
ADOBE, SAN JOSE RESEARCH SCIENTIST INTERN	May 2024 - Nov 2024
Multimodal generative models	Dr. Eli Shechtman Creative Intelligence Lab
PRIME VIDEOS & AMAZON STUDIOS, CULVER CITY APPLIED SCIENTIST INTERN	May 2022 - Aug 2022
Conditional video generative models	Naveen Nair, Ravi Teja Gadde Applied Science
SAMSUNG R&D INSTITUTE BANGALORE SOFTWARE ENGINEER (RESEARCH)	June 2018 - Aug 2020
Medical image analysis (prostate fusion biopsy)	Bhavya Ajani Health and Medical Equipment
SAMSUNG R&D INSTITUTE BANGALORE INDUSTRIAL INTERN	May 2017 - July 2017
Speech assistant's large utterance optimization	Vrajesh Sejpal Voice Intelligence
INDIAN STATISTICAL INSTITUTE, KOLKATA RESEARCH INTERN	May 2016 - Aug 2016
Quantum algorithms (k-SUM problem)	Prof. Subhamoy Maitra Applied Statistics Unit

EDUCATION

UNIVERSITY OF MARYLAND, COLLEGE PARK Aug 2020 - Present College Park, MD, USA	PH.D. IN COMPUTER SCIENCE (VISUAL GENERATIVE MODELS) Department of Computer Science Advisor - Prof. Abhinav Shrivastava GPA: 4.0 / 4.0
INDIAN INSTITUTE OF TECHNOLOGY GUWAHATI June 2014 - May 2018 Guwahati, India	BTECH IN COMPUTER SCIENCE AND ENGINEERING Department of Computer Science and Engineering Department Rank 5 (out of 88) GPA: 9.38 / 10.0

PAPERS

[Rethinking the Design Space of Vanilla Pixel Diffusion Transformers](#)

Soumik Mukhopadhyay, Long Mai, Abhinav Shrivastava, Aniruddha Mahapatra | Under Submission, 2026
Adaptive prediction parameterization and lightweight high-frequency refinement posttraining to improve Just Image Transformer (JiT).

[Scale Space Diffusion](#)

Soumik Mukhopadhyay, Prateksha Udhayan, Abhinav Shrivastava | CVPR, 2026
Why process noisy information-poor states at full resolution? Infusing scale spaces into diffusion for efficient pixel space diffusion.

[Selective Timestep Weighting and Advantage-Based Replay for Sample-Efficient Diffusion RLHF](#)

Eric Zhu, Abhinav Shrivastava, Soumik Mukhopadhyay | CVPRW (GCV, CVEU), 2026
Emphasizing information diffusion timesteps and trajectories during diffusion RLHF training provides up to 6x sample efficiency.

[Do text-free diffusion models learn discriminative visual representations?](#)

Soumik Mukhopadhyay, Matthew Gwilliam, Yosuke Yamaguchi, Vatsal Agarwal, Namitha Padmanabhan, Archana Swaminathan, Tianyi Zhou, Jun Ohya, Abhinav Shrivastava | ECCV, 2024
Diffusion models are self-supervised representation learners that can be used for various discriminative tasks as well.

[Diffusion Models Beat GANs on Image Classification](#)

Soumik Mukhopadhyay, Matthew Gwilliam, Yosuke Yamaguchi, Vatsal Agarwal, Namitha Padmanabhan, Archana Swaminathan, Srinidhi Hegde, Tianyi Zhou, Abhinav Shrivastava | Preprint only, 2023
Unconditional diffusion models' representations can be used for classification.

[Diff2Lip: Audio Conditioned Diffusion Models for Lip-Synchronization](#)

Soumik Mukhopadhyay, Saksham Suri, Ravi Teja Gadde, Abhinav Shrivastava | WACV, 2024
Diffusion models to synchronize face videos with arbitrary audio.

[Deep learning based needle tracking in prostate fusion biopsy](#)

Soumik Mukhopadhyay, Praful Mathur, Aditya Bharadwaj, Yuri Son, Jun-Sung Park, Srinivas Rao Kudavelly, Sangha Song, Hokyung Kang | SPIE Medical Imaging, 2021 (Best Poster Award in Image-Guided Procedures, Robotic Interventions, and Modeling)
Real-time localization of needle in an Ultrasound video of Prostate Biopsy and mapping to the 3D MRI space for analysis.

[Rigid and deformable corrections in real-time using deep learning for prostate fusion biopsy](#)

Aditya Bhardwaj, Jun-Sung Park, Soumik Mukhopadhyay, Sikander Sharda, Yuri Son, Bhavya Ajani, Srinivas Rao Kudavelly | SPIE Medical Imaging, 2020
Real-time MR-TRUS prostate registration: DL-based segmentation, elastix rigid registration, DL-based deformable correction.

PATENT

Methods and biopsy systems of detecting position of a needle during a biopsy procedure

Aditya Bharadwaj, Soumik Mukhopadhyay, Junsung Park, Srinivas Rao Kudavelly, Yuri Son
| Inventor on [Indian Patent No. 572418 \(Granted 2025\)](#)

¹ University of Maryland, College Park, ² Samsung R&D Institute Bangalore, ³ Indian Institute of Technology Guwahati, ⁴ Indian Statistical Institute Kolkata
⁵ Prime Videos & Amazon Studios, ⁶ Adobe

OTHER PROJECTS

1. Research Projects

Adobe ⁶	MULTIMODAL GENERATION	
	AUDIO-VISUAL JOINT GENERATION USING COMBINED TOKENIZATION Dr. Prem Seetharaman, Dr. Eli Shechtman, Adobe	May 2024 - Nov 2024
UMD ¹	FEW-VIEW 3D RECONSTRUCTION	
	SINGLE VIEW AUGMENTATION FOR IMPROVING DIFFUSION CLEANUP OF FEW-VIEW 3D Prof. Matthias Zwicker, Prof. Abhinav Shrivastava, Prof. Tom Goldstein, Dept. of Computer Science, University of Maryland	Sept 2025 - May 2026
	UNIFIED GENERATIVE REPRESENTATION LEARNING	
	GENERATIVE SELF-CONDITIONING FOR IMPROVED GENERATION Prof. Abhinav Shrivastava, Dept. of Computer Science, University of Maryland	Apr 2024 - Nov 2024
	VIDEO GENERATIVE MODELS	
	LIP-SYNCHRONIZATION IN HIGH-DEFINITION USING DIFFUSION MODELS Prof. Abhinav Shrivastava, Dept. of Computer Science, University of Maryland	Sept 2023 - Aug 2024
	IMPLICIT NEURAL REPRESENTATION	
	GENERATION IN NEURAL REPRESENTATION FOR VIDEOS (NeRV) SPACE Prof. Abhinav Shrivastava, Dept. of Computer Science, University of Maryland	Feb 2023 - Nov 2023
	MISCELLANEOUS	
	SPARSE FLOW BASED IN-BETWEENING FOR LINE DRAWINGS IN ANIMATION Prof. Abhinav Shrivastava, Dept. of Computer Science, University of Maryland	Aug 2022 - Oct 2022
	GENERALISED NEAREST NEIGHBOUR-BASED IMAGE GENERATION Prof. Abhinav Shrivastava, Dept. of Computer Science, University of Maryland	Feb 2022 - May 2022
	FACIAL PRIVACY IN GAN GENERATED IMAGES Prof. Abhinav Shrivastava, Dept. of Computer Science, University of Maryland	Mar 2021 - Dec 2021
	DETECTION OF FACIAL FORGEY USING GRAPH NEURAL NETWORKS Prof. Abhinav Shrivastava, Dept. of Computer Science, University of Maryland	Mar 2021 - Aug 2021
SRIB ²	MEDICAL IMAGE ANALYSIS	
	FETAL SKULL PREDICTIVE DETECTION/DATA AUGMENTATION USING DCGAN Nitin Singhal, Senior Chief Engineer, Samsung R&D Institute Bangalore	Nov 2018 – Feb 2018
	FOLLICLE DETECTION (FASTER-RCNN) & SEGMENTATION (UNET-BiLSTM) Nitin Singhal, Senior Chief Engineer, Samsung R&D Institute Bangalore	Sept 2018 – Nov 2018
IITG ³	COMPUTER VISION	
	END-TO-END DFT-BASED IMAGE DE-FENCING Prof. Arijit Sur, Associate Professor, Dept. of CSE, IIT Guwahati	Oct 2017 – Feb 2018

2. Course Projects

UMD ¹	VISION+TEXT	
	FINE-GRAINED IMAGE-TO-IMAGE EDITING FROM TEXT CAPTIONS Prof. Abhinav Shrivastava, Dept. of Computer Science, University of Maryland StyleGAN2 style attribute + BERT encoding for unsupervised image editing.	Sept 2020 - Dec 2020
	CONTINUAL LEARNING	
	ADVERSARIAL ROBUSTNESS AND CATASTROPHIC FORGETTING Prof. Soheil Feizi, Dept. of Computer Science, University of Maryland	Sept 2020 - Nov 2020

AWARDS/ACHIEVEMENTS

2024	Kulkarni Foundation Summer Research Fellowship (Declined)
2021 & 2023	CS Summer Research Fellowship, UMD
2020	Dean’s Fellowship 2020, UMD
2019	Paper Selected for presentation at Samsung Best Paper Award (among top 8)
2019	Samsung Spot Award for Successful Implementation of Biopsy Needle Detection on Live Ultrasound Data
2019	Samsung Professional Level Software Certification
2013-14	Kishore Vaigyanik Protsahan Yojana (KVPY) Fellowship
2014	Among the top 0.3% of 1.3 million in the Joint Entrance Examination (JEE), India
2014	Letter of Appreciation from the HRD Minister for being in the top 1 percentile of CBSE (in English)

COURSEWORK

GRADUATE	• Advanced Techniques in Visual Learning and Recognition • Foundations of Deep Learning • AI and Existential Threats to Civilization • Advanced Numerical Optimization • Just Machine Learning • Computer Processing of Pictorial Information • Machine Learning Paradigms
UNDERGRADUATE	• Computer Vision using Machine Learning • Speech Processing • Artificial Intelligence • Parallel Algorithms • Computational Theory • Algorithmic Game Theory • Number Theory and Cryptography • Operating Systems • Software Engineering • Database Management System • Computer Networks • Probability Theory • Algorithms • Data Structures • Discrete Mathematics • Linear Algebra
ONLINE (Coursera)	1. Fundamentals of Digital Image and Video Processing by Aggelos K. Katsaggelos 2. Deep Learning Specialization by Andrew Ng

SKILLS

PROGRAMMING LANGUAGES	Python • C • C++
TOOLS, FRAMEWORKS, AND LIBRARIES	Pytorch • Keras • TensorFlow • ITK • Matlab*

* Elementary Proficiency

¹ University of Maryland, College Park, ² Samsung R&D Institute Bangalore, ³ Indian Institute of Technology Guwahati, ⁴ Indian Statistical Institute Kolkata
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